

TABLE 2 Genotypes of 21 sporadic BWS patients for nine 11p15.5 RFLPs

	HRAS1 <i>Bam</i> HI	TH <i>Taq</i> I	INS <i>Rsa</i> I	INS <i>Hind</i> III	IGF2 <i>Sac</i> I	IGF2 <i>Av</i> all	D11S12 <i>Msp</i> I	HBB <i>Av</i> all	HBBP <i>Hinc</i> II
BW1P	1/1	2/2	2/2	1/1	1/2	1/2	2/2	1/2	2/3
BW2P	1/1	2/2	2/2	1/2	1/2	1/2	2/2	1/2	2/3
BW3P	1/2	2/2	2/2	2/2	2/2	1/1	1/2	2/2	1/1
BW4P	1/2	2/2	2/2	2/2	2/2	1/1	2/2	2/2	1/1
BW5P	1/1	2/2	1/2	2/2	2/2	1/1	2/2	2/2	1/1
BW6P	1/2	2/2	2/2	2/2	2/2	1/1	1/2	1/2	1/1
BW7P	1/2	2/2	2/2	1/1	2/2	1/1	2/2	2/2	1/1
BW8P	2/2	2/2	1/1	1/1	1/2	1/2	1/2	2/2	1/2
BW9P	1/2	1/2	2/2	2/2	2/2	1/1	1/2	1/2	1/1
BW10P	1/2	2/2	1/1	1/1	1/2	1/1	2/2	1/2	1/1
BW11P	2/2	2/2	1/1	1/1	2/2	1/1	2/2	2/2	1/1
BW12P	2/2	2/2	1/1	1/2	1/1	2/2	1/2	1/1	2/2
BW13P	1/2	2/2	1/1	1/2	2/2	1/1	1/2	2/2	1/2
BW14P	1/2	2/2	1/1	2/2	1/2	1/1	1/2	2/2	1/1
BW15P	2/2	1/2	2/2	2/2	2/2	1/1	1/1	2/2	1/1
BW16P	1/2	2/2	2/2	1/2	1/2	1/2	2/2	1/2	1/3
BW17P	2/2	2/2	1/2	1/2	2/2	2/2	2/2	2/2	1/3
BW18P	2/2	2/2	1/2	2/2	2/2	1/1	2/2	2/2	1/1
BW19P	2/2	1/2	2/2	2/2	2/2	1/1	1/2	2/2	1/3
BW20P	1/2	1/2	2/2	2/2	2/2	1/1	1/2	2/2	1/2
BW21P	2/2	2/2	1/1	1/1	2/2	1/1	2/2	2/2	2/2
F_{BWS}	0.52	0.80	0.85	0.76	0.71	0.80	0.57	0.66	0.57
F_C	—	0.78 (28/36)	0.52 (19/36)	0.51 (18/35)	0.51 (18/35)	0.58 (21/36)	—	—	—
$F_{C/HGM10}$	0.52	0.73	0.52	0.45	0.55	0.62	0.73	0.68	0.51

F_{BWS} , Frequency of homozygous individuals in the BWS population; F_C , frequency of homozygous individuals in the control population. The numbers in brackets indicate the number of homozygous individuals versus the total number of individuals tested. $F_{C/HGM10}$, Frequency of homozygotes as from HGM 10 (ref. 16); P , probability that the random distribution of the alleles is responsible for the higher homozygosity in patients than in controls. For the four contiguous loci (INS/*Rsa*I, INS/*Hind*III, IGF2/*Sac*I, and IGF2/*Av*all) taken together, the frequency of homozygotes was 0.48 (10/21) for BWS patients and 0.18 (6/34) for controls. Methods and probes were as described in Table 1. The test used is a simple one-tailed χ^2 homogeneity test (valid since all the expected values are greater than 6); $\chi^2 = 5.65$, $P < 0.01$. The one-tailed test was used because in the hypothesis of isodisomy the frequency of homozygotes is expected to be higher than in controls.

syndromes: the Prader-Willi syndrome with a lack of 15q11-q13 paternal alleles¹⁴, and the Angelman syndrome with a lack of 15q11-q13 maternal alleles¹⁵. Taken together the lack of maternal alleles in the three sporadic BWS cases reported here, paternal duplication in trisomic BWS patients, retention of paternal alleles in tumours, and higher penetrance in individuals born to female carriers in familial BWS corroborate the involvement of genomic imprinting. Thus we conclude that an unbalanced dosage of maternal and paternal alleles is a common feature for the different aetiological forms of BWS and associated tumours. □

Embryological and molecular investigations of parental imprinting on mouse chromosome 7

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MOUSE embryos with duplications of whole maternal (parthenogenetic and gynogenetic) or paternal (androgenetic) genomes show reciprocal phenotypes and do not develop to term^{1,2}. Genetic complementation has identified the distal region of chromosome 7 (Chr 7) as one of the regions for which both a maternal and paternal chromosome copy are essential for normal development, presumably because of the presence of imprinted genes whose expression is dependent on their parental origin^{3,4}. Embryos with the maternal duplication and paternal deficiency of distal Chr 7 are growth retarded and die around day 16 of gestation; the reciprocal paternal duplication embryos die at an unidentified earlier stage⁴. We report here the incorporation of cells with the paternal duplication into chimaeras, resulting in a striking growth enhancement of the embryos. One gene located on mouse distal Chr 7 (ref. 5) is the insulin-like growth factor 2 (*Igf2*) gene, an embryonic mitogen⁶. In embryos with the maternal duplication of distal Chr 7, the two maternal alleles of the *Igf2* gene are repressed. The presence of two paternal alleles of this gene in many cells is probably responsible for the growth enhancement observed in chimaeras. We propose that there are other imprinted genes in

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this Chr 7 region. We also compare the imprinting of this sub-genomic region with phenotypes resulting from the duplication of the whole parental genome in parthenogenones and androgenones.

We investigated the developmental potential of cells carrying the paternal duplication of the region distal to the T(7B3;15C)9H translocation breakpoint on Chr 7 (ref. 7) in chimaeras. Blastocysts carrying the paternal duplication of distal Chr 7 (PatDi7) were expected at a frequency of about 17% (ref. 8). Two markers were used to identify the experimental cells: glucose phosphate isomerase-1 (*Gpi-1*), and albino (*c*) which is located on distal Chr 7. Inner-cell masses (ICMs) were released from putative PatDi7 blastocysts and injected into

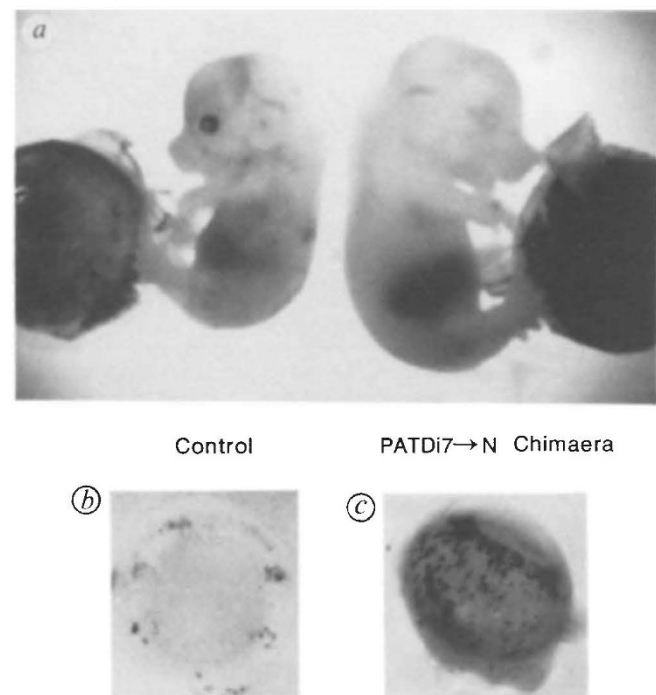


FIG. 1 *a*, Comparison of (e15) PatDi7→N chimaera (right) embryo and control-manipulated but nonchimaeric littermate (left) showing growth enhancement in the chimaera due to the presence of cells carrying the paternal duplication of the distal region of Chr 7. The chimaera was about 50% larger by weight than controls. *b*, Eye from e15 PatDi7→N chimaera in *a*. The pigmented cells are derived from the host component and the *c/c* cells from the donor PatDi7 ICM. The amount of pigment reflects the extent of chimaerism as assayed by GPI-1 isozyme analysis. *c*, Eye from e17 PatDi7→N chimaera with pigmented and nonpigmented components as described in *b*. This embryo was about 55% larger by weight than the controls. METHODS. Construction of chimaeras: e4 blastocysts (*Gpi-1^a/Gpi-1^a*) were isolated from uteri of T9H+/++♀×T9Hc/+♂ intercrosses and cultured *in vitro* for 2 days²³. These blastocysts included the genetically marked and balanced embryos carrying the paternal duplication occurring at a frequency of about 17%⁸. Many of the embryos from the intercross were arrested around the four-cell stage due to the many genetically unbalanced gametes produced. This further reduced the number of blastocysts which provided donor ICMs. ICMs were isolated by immunosurgery²⁴ and microinjected into expanded normal host blastocysts (CBA×C57B1/6J)_{F₁} (*Gpi-1^b/Gpi-1^b*) using methods described previously^{23,25,26} with the following modifications (S.C.B. & R. Fundele, unpublished results); injection pipettes were bevelled, with an internal diameter of 22 µm and ICMs were briefly stored in medium supplemented with cytochalasin B (5 µg ml⁻¹) and Nocodazole (1.5 µg ml⁻¹) to allow them to be distorted without damage to the cells. The manipulated embryos were then transferred into the uteri of pseudopregnant recipients 3 days *post coitum*. Embryos were removed at stages e15–17.5, dissected and analysed for GPI-1 isozyme type for chimaerism²³. Chimaeric embryos were GPI-1A+GPI-1B type and in the PatDi7→N chimaeras also had albinism in the eye indicative of two copies of paternal *c* alleles in the pigment cells derived from the donor ICM. Growth enhancement always correlated with albinism in the eye.

normal host blastocysts (N) to generate chimaeras. Three PatDi7→N chimaeras were obtained and exhibited a striking phenotype; a substantial size increase (Fig. 1*a* and Table 1). For an e15 PatDi7→N embryo the increase was 51% with the PatDi7 cells present at about 75% in 10 tissues analysed (Fig. 1*b* and Table 1). An e15.5 PatDi7→N embryo was 35% larger than controls with 50% contribution of PatDi7 cells in limbs which were the only tissues assayed for GPI; the remaining embryo was used for northern blot analysis. The 17-day-old PatDi7→N embryo was 55% larger with nearly equal contributions to the 14 tissues analysed (Fig. 1*c* and Table 1). These sizes were compared with both manipulated and nonmanipulated embryos carrying normal ICMs. The results demonstrate that the PatDi7 cells were 'rescueable' in chimaeras and that a large contribution from PatDi7 cells was responsible for enhanced growth.

The maternal duplication of distal Chr 7 produces a less severe phenotype with embryos dying around day 16. ICMs from blastocysts, with the maternal duplication of distal Chr 7 (MatDi7), also expected at the low frequency of about 17% (ref. 8) were introduced into normal host blastocysts. These embryos were allowed to develop to term. Two of these were MatDi7→N chimaeras, as determined by coat colour and GPI-1 analysis. The contribution of MatDi7 cells was approximately 25–67% in all 28 tissues analysed in one 2-week-old mouse and in 18 tissues analysed in a 6-month-old mouse without major differences in the relative tissue distributions between the two animals. Importantly, the body weights of the MatDi7→N chimaeras at or after birth did not differ significantly from those of the chimaeric or nonchimaeric controls (see legend to Fig. 2).

To identify imprinted genes on distal Chr 7, nonchimaeric MatDi7 embryos, which can develop to day 16 were used because, although these are roughly 50% smaller than controls⁴ (Fig. 2*a*), there were no obvious histological malformations (not shown). Expression of genes which map to distal Chr 7 was investigated by northern blot analysis. One of these genes is *Igf2* (ref. 5) which is an embryonal mitogen⁶ widely expressed in several embryonic and extraembryonic tissues in the mouse⁹. The expression of *Igf2* was assayed because of the growth phenotypes described here. *Igf2* expression is virtually undetectable in the e13–e15 MatDi7 embryos and placentae compared with controls (Fig. 2*b*), although low expression from maternal alleles is not excluded which may be confined to a subset of cells (see ref. 10). The e15.5 PatDi7→N embryo chimaera also showed an increase in *Igf2* expression of about 31% as assayed by a series of densitometric scans of northern blots (not shown). In the e15 hairpin-tail mutant embryos (T^{hp}), which were used as controls because, like the MatDi7 embryos, they also die at day 16 (see legend to Fig. 2), *Igf2* expression was normal (Fig. 2*b*). These data are consistent with RNase protection data derived from targeted mutagenesis of *Igf2* (ref. 10). The *Igf2* receptor gene is also imprinted but with the maternal allele being active and the paternal allele being inactive¹¹; the opposite of the ligand. These findings suggest a critical role for these imprinted genes in controlling normal mammalian development.

It is of interest to compare these chimaeric phenotypes with those containing either androgenetic (AG) or parthenogenetic (PG) cells in the presence of normal cells (N). The AG→N chimaeras showed a comparable size increase to that of the PatDi7→N chimaeras¹² whereas the PG→N chimaeras were smaller by a similar proportion¹³. Indeed preliminary results show that *Igf2* expression is substantially higher in AG than in PG conceptuses (R. Ohlsson, A. Glaser, A.F.-S., S.C.B. and M.A.S., manuscript in preparation). The size of MatDi7→N chimaeras was comparable to the controls, which could be attributed to the overall dosage of *Igf2* reaching a threshold in the presence of normal cells. This is consistent with the proposed autocrine-paracrine action of *Igf2*^{6,14}. Hence, the growth retardation in PG→N chimaeras is probably due to the cumulative effects of other imprinted genes in PG cells involved in

growth regulation. The distribution of AG and PG cells in chimaeras is non-uniform and reciprocal, unlike that of the MatDi7 and PatDi7 cells. AG cells contribute disproportionately to tissues of mesodermal origin such as the skeletal muscle, heart and kidney and to ectodermally derived tissues such as the brain and epidermis^{12,15}. The contribution of PG cells in chimaeras is most consistently in the brain, epidermis and germ

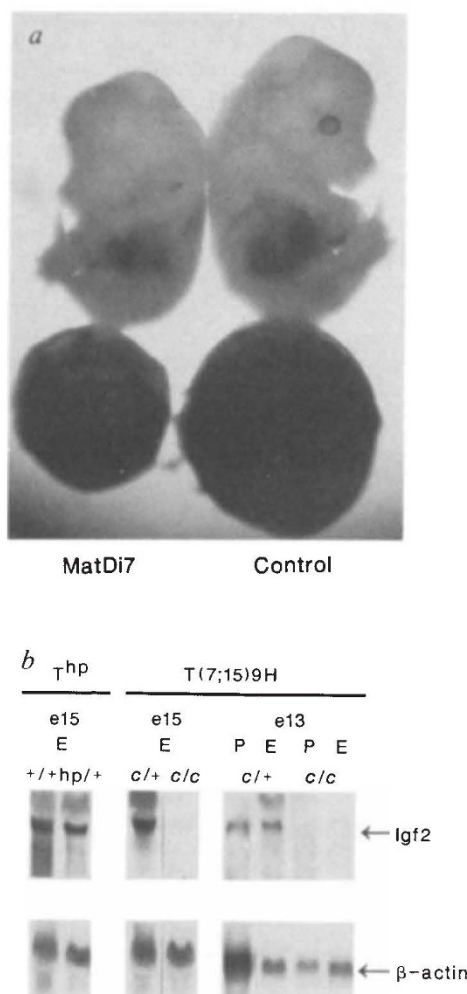


FIG. 2 *a* Comparison of e15 MatDi7 (left) and control embryo and placenta derived from the T9Hc/+c $\times$ T9H+/++ δ intercross⁴ illustrating the growth retardation of both the embryo and placenta in the presence of two maternal copies of the distal region of Chr 7. The MatDi7 embryo lacks pigment in the eye due to the presence of the two maternal *c* alleles. Weights of two liveborn MatDi7 $\rightarrow$ N chimaeras were 5.0 g and 4.0 g compared with 5.7 g and 4.1 g, respectively, of concurrent controls indicating no significant size difference. *b*, Igf2 expression analysis of experimental and control e13 and e15 embryos (E) and placentae (P). T9H RNA was from T9Hc/+c $\times$ T9H+/++ δ . *c/c*, Maternal duplication for distal Chr 7; *c/+*, nonalbino littermates (see *a*). T^{hp} RNA was from T^{hp} ins/+ (CBA $\times$ C57B1/6J)F₁ $\times$ (CBA $\times$ C56B1/6J)F₁ δ ²⁷. In this intercross, maternal inheritance of the mutation results in a mutant phenotype whereas the reciprocal intercross, where the mutation is paternally inherited, does not^{11,27}. Mutant embryos were distinguished from normal littermates by their mutant tails and the unambiguous presence of oedema. +/+, Control littermate; hp/+, maternally inherited hairpin-tail mutant embryos. The Igf2 probe is derived from exon 6 (ref. 5) hybridizing to a major transcript of ~4.8 kilobases. β -actin is shown as a qualitative and quantitative control and illustrates that actin is transcribed in the absence of Igf2 in both the *c/c* embryo and placenta. METHODS. Embryos were killed at e13–e15 and RNA was isolated separately from embryos and placentae using methods described previously²⁸. RNA (10–20 μ g) was separated on formaldehyde gels, transferred to nylon membranes and hybridized as shown with ³²P-radiolabelled probes²⁹.

TABLE 1 Analysis of embryos

Embryo age (days)	No. of litters	Embryo	n	GPI	Mean weight $\pm$ s.d. E(mg)	P(mg)
e15	1	Unoperated	2	B	258 $\pm$ 44*	89 $\pm$ 12
		Unchimaeric	2	B	230 $\pm$ 18*	81 $\pm$ 5
		PatDi7 $\rightarrow$ N	1	A + B	369 (51%)	117
e15.5	2	Unoperated	5	B	348 $\pm$ 32	99 $\pm$ 12
		Unchimaeric	4	B	326 $\pm$ 13	81 $\pm$ 12
		Chimaeric	1	A + B	329	99
		PatDi7 $\rightarrow$ N	1	A + B	450 (35%)	104
e16	4	Unoperated	12	B	402 $\pm$ 68	98 $\pm$ 12
		Unchimaeric	10	B	398 $\pm$ 42	94 $\pm$ 15
		Chimaeric	1	A + B	406	113
e17	5	Operated	11	B	615 $\pm$ 79	108 $\pm$ 10
		Unchimaeric	14	B	522 $\pm$ 77	116 $\pm$ 27
		Chimaeric	9	A + B	599 $\pm$ 88	104 $\pm$ 20
		PatDi7 $\rightarrow$ N	1	A + B	869 (55%)	111
e17.5	2	Unchimaeric	5	B	640 $\pm$ 122	109 $\pm$ 14
		Chimaeric	4	A + B	791 $\pm$ 174	113 $\pm$ 15

A total of 83 embryos were recovered of 117 transferred to pseudopregnant recipients; 53 manipulated embryos of 80 transferred were recovered; 3 recipients failed to become pregnant. e1 is the day of vaginal-plug appearance. E, Embryo; P, placenta; s.d., standard deviation for $n > 2$; n , number of embryos recovered; N, normal fertilized component of chimaera. The percentage in parenthesis indicates the growth enhancement of the PatDi7 $\rightarrow$ N chimaera compared with all concurrent controls. GPI, the isozyme type denoted A and/or B. The following 14 tissues were dissected from each manipulated embryo and assayed for chimaerism by GPI-1 typing: brain, eyes, tongue, heart, lung, thymus, forelimb, hindlimb, gut, pancreas, liver, kidneys, gonads. For some embryos placenta, blood and skin were included. Chimaeras contained roughly uniform contributions of experimental and host cells to all tissues except for placenta and blood where there were very few experimental cells. This is expected for placenta which is predominantly host-derived. Only the limbs of the e15.5 PatDi7 $\rightarrow$ N were assayed for GPI-1.

* These values are means $\pm$ difference.

cells^{12,13,16}. By contrast, MatDi7 cells are detected in skeletal muscle to which PG cells rarely make a contribution, presumably due to the action of other imprinted genes in PG cells.

We believe that imprinted genes in addition to *Igf2* are present on distal Chr 7. First, targeted mutagenesis of *Igf2* followed by the transmission of the intact maternal allele resulted in mice that, although growth retarded, were live and fertile¹⁷ whereas the MatDi7 embryos die at around day 16. Second, duplications of a region more distal to the T(7; 15)9H breakpoint may also produce embryos of a less severe phenotype⁴. The early demise of the unchimaeric PatDi7 embryos may be a result of two active paternal alleles of *Igf2* or due to the action of other imprinted genes.

Human Chr 11p15, which shares syntenic homology with mouse distal Chr 7, may also be imprinted. The paternal duplication of the region seems to be associated with enhanced growth dysmorphologies linked to the Beckwith–Wiedemann syndrome and also to embryonal tumours of mesodermally derived tissues including rhabdomyosarcoma and Wilms' tumour^{18–20}. The expression of *Igf2* is elevated in these tumours as well as tumours of the trophoblast²¹, a site of normal *Igf2* expression. A cloned sequence with homology to the imprinted region of human Chr 15 associated with the Prader–Willi/Angelman syndromes has been mapped to mouse Chr 7²². Detailed investigations in the mouse, particularly of the paternal duplication of distal Chr 7 as seen in the enhanced growth of PatDi7 $\rightarrow$ N embryos, may aid our understanding of the genesis of these human genetic disorders.

We propose that this parental duplication–deficiency system provides a unique and powerful model to investigate the mechanism of imprinting. This is because both imprinted alleles of a gene are of the same parental origin and therefore likely

to have the same molecular characteristics, unlike in normal cells where the two parental alleles will have a different imprint. We have thus begun a detailed analysis of this chromosomal region using cells carrying the maternal duplication. These studies could provide an understanding of the chromatin structure and DNA modifications responsible for the differential parental gene activities which underlie chromosomal imprinting. *Note added in proof.* While this paper was in press an additional gene on distal Chr 7, H19, was shown to be imprinted³⁰ supporting our proposal that there is at least one other imprinted gene in this region. □

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report that purified NodRm-1 and a related compound, Ac-NodRm-1, at concentrations in the micromolar–nanomolar range, elicit cortical cell divisions and the formation of genuine nodules on aseptically grown seedlings of alfalfa. Chemical modifications of NodRm-1, such as the removal of the sulphate group, reduction of the terminal sugar or hydrogenation of the acyl chain result in a strong decrease in the morphogenic activity. A highly specific prokaryotic lipo-oligosaccharide signal is thus able to trigger a genuine organogenesis in a higher plant.

The early steps in the development of the alfalfa–*R. meliloti* symbiosis involve initiation by the bacterium of the curling of host root hairs and the triggering of cortical cell division leading to nodule organogenesis¹. Mutations in the *R. meliloti nodABC* genes abolish all these processes and also prevent the production of the NodRm-1 factor^{1,4,8}. The *nodABC* operon is expressed in the infection threads close to which cortical cells are dividing (Fig. 1a). Thus NodRm-1 might be involved in the induction not only of root hair deformation but also of cortical cell divisions.

NodRm-1 is obtained from the culture supernatant of *R. meliloti* strains that have been genetically engineered to overproduce the signal⁴. A related factor, Ac-NodRm-1, which is *O*-acetylated at carbon 6 of the terminal non-reducing sugar (Fig. 2), is also found in amounts that vary with each culture⁷. As NodRm-1 and Ac-NodRm-1 are difficult to separate completely, the *O*-acetyl group⁷ was removed in order to use a defined, purified molecule, NodRm-1, in the following experiments.

Microscopy studies of the root system of alfalfa seedlings grown for 6, 9 or 12 days in the presence of NodRm-1 showed that at a concentration of 10^{-9} M, NodRm-1 induced discrete foci of cell divisions in the inner cortex. No such foci could be observed for untreated control seedlings. At a higher concentration (10^{-7}) cell divisions either occurred as discrete foci (Fig. 1b) or extended to larger zones of the root including the whole cortex (Fig. 1f, g).

Alfalfa seedlings were grown aseptically for 4 weeks in the presence of various concentrations of NodRm-1. Root deformations were observed which were distinguishable from the cone-shaped protuberances of lateral roots, and varied in appearance from bumps to round, elongated or multilobate structures (Fig. 1c). Detailed microscopical studies defined their ontogeny, anatomy and histology, to determine on the basis of defined criteria^{8,26} whether they are nodules, roots or intermediary structures. Light microscopy showed that these deformations had an early developmental pattern characteristic of rhizobia-induced nodules. The meristems originated from the activity of foci of dividing cortical cells (Fig. 1d, g). In emerging deformations, early cortical cell divisions were followed by an activation of the underlying endodermis and vascular bundles (Fig. 1h). The newly formed bundles developed from at least two, but often four, distinct sites close to the root vascular system (Fig. 1h–j). Later, most of the deformations protruding from the root surface had anatomical and histological features of genuine rhizobia-induced nodules, such as apical meristems and peripheral vascular bundles and endodermis (Fig. 1i, j). These anatomical characteristics were fundamentally different to those of equally-sized developing lateral roots which exhibited differentiation of a single central vasculature surrounded by endodermis (compare Fig. 1d and e, and the root deformation and the lateral root in Fig. 1j). The NodRm-1-induced nodules were bacteria-free because (1) we could not detect colony-forming units when the crushed root-systems of NodRm-1-treated plants were streaked on agar plates; and (2) we could not detect any bacteria by light and electron microscopy of these structures (data not shown). Histological studies of all such root deformations showed that more than 99% of the structures scored on morphological criteria (a nonconical shape) were genuine nodules. We therefore used visual scoring to assay nodule formation in the quantitative experiments.

Sulphated lipo-oligosaccharide signals of *Rhizobium meliloti* elicit root nodule organogenesis in alfalfa

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RHIZOBIUM meliloti is a symbiotic bacterium that elicits the morphogenesis of nitrogen-fixing nodules, specific organs on the roots of alfalfa (*Medicago sativa*)¹. In *R. meliloti* a series of nodulation (*nod*) genes have been identified which are involved in root-hair curling and infection and in nodule formation¹. The *nodABC* genes, common to all *Rhizobium* sp., and the host-range *nodH* and *nodPQ* genes are involved in the production of an excreted root-hair deforming factor, NodRm-1, which is a sulphated and acylated glucosamine oligosaccharide^{2–7}. Here we